A PROJECT TITLE

ON

Portfolio Risk Analysis Using Principal Component Analysis (PCA)

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

by

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DEPARTMENT OF CSE

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1. ABSTARCT

Portfolio Risk Analysis Using Principal Component Analysis (PCA) is a machine learning project that analyzes the risk associated with a portfolio of stocks using historical market data. The system retrieves financial data, preprocesses it, and applies Principal Component Analysis (PCA) to identify the key factors that influence portfolio performance while reducing data complexity.

The project provides insights into asset correlations, explained variance, and portfolio diversification through interactive visualizations and statistical analysis. By reducing high-dimensional financial data into a few principal components, the system helps users understand the major sources of portfolio risk and supports informed investment analysis.

The proposed system utilizes Python and machine learning libraries to perform data preprocessing, dimensionality reduction, and visualization. To make the analysis more practical, the project can retrieve the latest historical stock data from freely available financial sources, enabling dynamic and up-to-date portfolio analysis. This project demonstrates the real-world application of unsupervised machine learning techniques in financial risk assessment and data-driven decision-making.

2. INTRODUCTION:

Portfolio Risk Analysis Using Principal Component Analysis (PCA) is a machine learning-based system designed to analyze and understand the risk associated with a portfolio of stocks. Financial markets generate large amounts of data, making it difficult to identify the major factors that influence stock prices and portfolio performance. This project uses historical stock market data to study relationships between different assets and identify the key sources of portfolio risk.

The system preprocesses financial data and applies Principal Component Analysis (PCA), an unsupervised machine learning technique used for dimensionality reduction. PCA transforms a large number of correlated financial variables into a smaller set of principal components while retaining the most important information. This helps in identifying dominant patterns, asset correlations, and major factors affecting portfolio performance.

The project also provides statistical analysis and interactive visualizations such as correlation matrices, explained variance plots, and principal component representations. These results help users understand portfolio diversification and risk concentration. Python and its data science libraries are used to collect, process, analyze, and visualize financial data. The system can use freely available historical market data, making the analysis dynamic and useful for practical financial decision-making

3. SOFTWARE REQUIREMENTS

1. Operating System
- Windows 10/11, Linux, or macOS

2. Programming Language
- Python 3.9 or above
- python

3. Development Environment
- Jupyter Notebook or Visual Studio Code

4. Python Libraries
- NumPy – Numerical calculations and array operations
- Pandas – Data preprocessing and handling financial datasets
- Scikit-learn – PCA and machine learning operations
- Matplotlib – Data visualization
- Seaborn – Statistical and correlation visualizations
- yFinance – Retrieving historical stock market data

Signature of the Guide **Course Instructor**